Exp.no : 01 **DATA ANALYSATION**

a) AIM:

Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib for visualization.

PYTHON CODE:

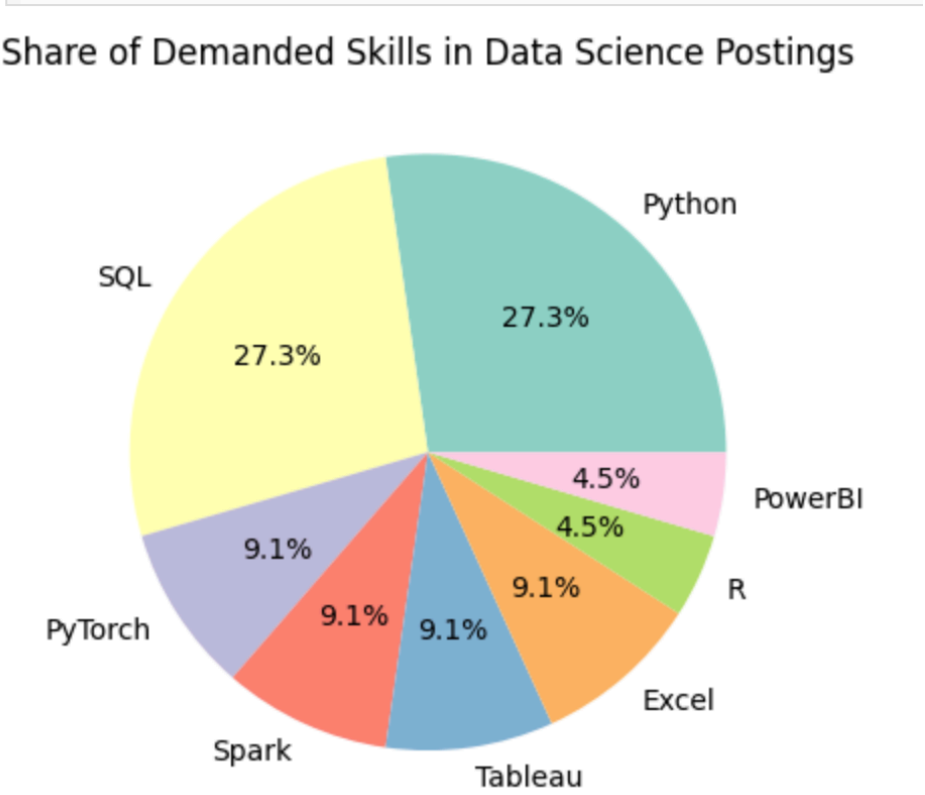
```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Job_Title': [
        'Data Scientist', 'Data Engineer', 'Data Analyst',
        'ML Engineer', 'Data Scientist', 'Data Analyst',
        'Data Engineer', 'Data Scientist'
    ],
    'Skills': [
        'Python, SQL, PyTorch',
        'Python, SQL, Spark',
        'SQL, Tableau, Excel',
        'Python, PyTorch',
        'Python, R, SQL',
        'Excel, SQL, PowerBI',
        'Python, Spark',
        'Python, SQL, Tableau'
    ]
}

df = pd.DataFrame(data)
skills_series = df['Skills'].str.split(', ').explode()
skill_counts = skills_series.value_counts()
plt.pie(
    skill_counts.values,
    labels=skill_counts.index,
    autopct='%1.1f%%',
    colors=plt.cm.Set3.colors
)

plt.title('Share of Demanded Skills in Data Science
Postings')

plt.show()
```

OUTPUT:



RESULT:

The code is executed successfully.

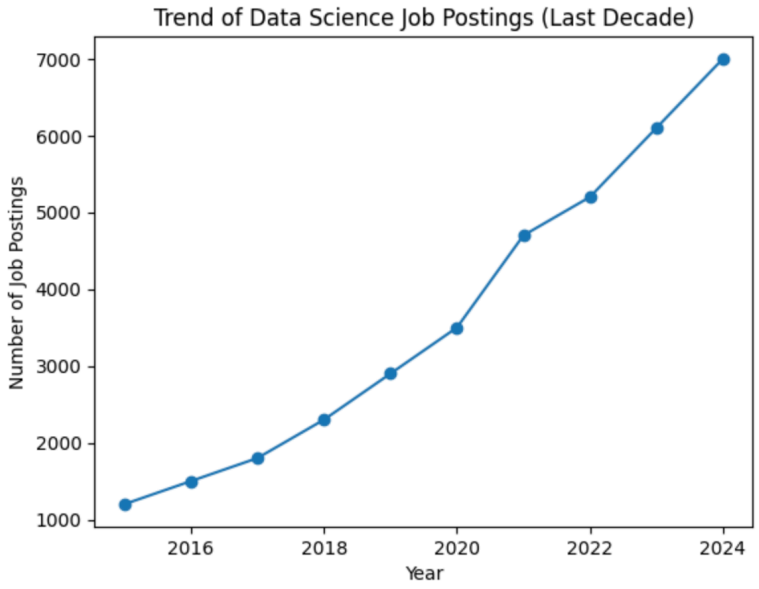
b) AIM:

Analyzeand visualize the distribution of various data science roles (Data Analyst,Data Engineer, Data Scientist, etc.) from a dataset.

PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022,
2023, 2024],
    'Job_Postings': [1200, 1500, 1800, 2300, 2900, 3500,
4700, 5200, 6100, 7000]
}
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job_Postings'], marker='o')
plt.title('Trend of Data Science Job Postings (Last Decade)')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

OUTPUT:



RESULT:

The code is executed successfully.

C) AIM:

Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

PYTHON CODE:

```
import pandas as pd

# Structured Data
structured = pd.DataFrame({
    "Name": ["Alice", "Bob"],
    "Age": [20, 21]
})
print(structured)

# Semi-Structured Data
semi = {
    "Name": "Alice",
    "Age": 20
}
print(semi)

# Unstructured Data
text = "Artificial Intelligence is changing the world."
print(text)
```

OUTPUT:

```
Name  Age
0  Alice   20
1    Bob   21
{'Name': 'Alice', 'Age': 20}
Artificial Intelligence is changing the world.
```

RESULT:

The code is executed successfully.

D) AIM:

Conduct an experiment to encrypt and decrypt given sensitive data using the cryptography library fernet.

PYTHON CODE:

```
from cryptography.fernet import Fernet

# Generate a key
key = Fernet.generate_key()
f = Fernet(key)

# Original data
data = "PIN-8#290"
print("Original Data:", data)

# Encrypt the data
encrypted = f.encrypt(data.encode())
print("\nEncrypted Data:", encrypted)

# Decrypt the data
decrypted = f.decrypt(encrypted).decode()
print("\nDecrypted Data:", decrypted)
```

OUTPUT:

```
Original Data: My ATM PIN is 1234

Encrypted Data: b'gAAAAABqchkp4FK4b5PjhhF2ZIW6jbLJ7zET9w3-
U57owpV_ICEw63YvtkpO5yTotmcgb5imtJCSlwyDQLmXXi8ABkVvkx5xJfmL
dVcfgBReBxPNikJbxlA='
```

RESULT:

The code is executed successfully.